

An Explainable AI Model for IoMT Network Attack Detection

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Abstract

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1 Introduction

Smart and connected medical devices are increasingly integrated into the delivery of healthcare services, leading to the development of the Internet of Medical Things (IoMT), a subset of the Internet of Things (IoT) encompassing medical equipment and devices. IoMT technology offers many advantages by enabling continuous monitoring of patients' health. This assists healthcare professionals in making remote diagnoses and better informing their clinical decisions; for example, data collected from IoMT devices has been used to predict patients' risk of heart disease [1]. With the growing digitalization of healthcare systems, there has been a significant increase in cyberattacks targeting medical infrastructure. According to the HIPAA Journals' report, there have been 546 healthcare breaches that affect more than 500 people so far in 2025 [2]. This, coupled with the sensitive nature of medical data, raises major concerns about patients' privacy, where patients' personal data may be exposed, as well as potential concerns about patients' safety, given that IoMT devices are used to deliver treatments.

Recent research has shown that Machine Learning (ML) can be effectively used to detect network attacks by analyzing network traffic. Compared to traditional network attack detection methods, such as rule-based systems, ML techniques offer significant improvements by recognising complex patterns in network traffic that were previously not possible. However, the problem with complex ML techniques is their lack of interpretability in decision-making; in the context of healthcare systems, this interpretability is crucial, as medical staff and patients need to trust that IoMT networks are secure. To address the lack of interpretable ML models, this project will develop an Explainable Artificial Intelligence (XAI) model to detect network attacks in IoMT systems. This ensures that users of the system can understand and trust the model while still achieving high detection accuracy.

This project seeks to design and implement an XAI model that can detect network attacks in an IoMT system, whilst maintaining interpretability in its detections. The proposed model will be trained and evaluated using the CIC2024 IoMT dataset, a publicly available dataset synthesised by the Canadian Institute for Cybersecurity [3]. This dataset has been specifically designed to develop and benchmark network attack detection models. It simulates a number of IoMT devices and a wide variety of both benign and malicious network traffic.

The primary objectives of this project are: (1) to review current literature on AI-based network attack detection within IoT and IoMT networks; (2) to develop an accurate and interpretable attack detection model and select a suitable XAI technique; (3) Evaluate the models performance using the CIC2024 IoMT dataset; (4) Assess the models interpretability and its suitability for cybersecurity in a healthcare context.

2 Literature Review

2.1 IoMT Architecture and Security Concerns

IoMT systems are typically divided into three primary layers: perception, network, and applications. [4].

In the perception layer, we have the actual medical sensors and wearable devices that collect information from patients, such as tracking cardiovascular and heart health, temperature, and sleep patterns. This layer faces potential vulnerabilities, with the potential for tampering and false data collection, which could ultimately lead to incorrect diagnoses based on incorrect data.

The network layer enables these devices to communicate with other devices in the IoMT network using protocols such as Wi-Fi and Bluetooth. This layer is susceptible to attacks such as DoS and DDoS, which can lead to disruptions in treatment and pose an immediate risk to patients' safety when IoMT devices are unable to operate. Man-in-the-middle attacks at this layer can also lead to data breaches, compromising the confidentiality of sensitive patient and medical staff data.

Moving to the application layer, we find software vulnerabilities where cyber threats, such as phishing and ransomware, can lead to interruptions in medical care and further data breaches. Across

these layers, there are three clear points of risk for IoMT technology: patients' data privacy, patients' safety, and disruption to medical systems.

The variety of areas that are vulnerable in these systems highlights the need for a more rigorous and trustworthy cybersecurity system to reduce the number of successful attacks. Effective Intrusion Detection System (IDS) is an essential tool for preventing the wide variety of attacks that can IoMT networks face.

2.2 ML-Driven IDS in IoMT Environments

Traditional Intrusion Detection Systems (IDS) such as signature based methods are not well-suited to the field of IoT networks, this has led to research in recent years primarily being focused on hybrid Machine Learning (ML) methods for detecting network attacks, including Random Forest and SVM [5]. These ML-based systems aim to analyse large amounts of network traffic and distinguish between benign and malicious activity. In this section, I will focus on reviewing the most well-documented models used and how their performance is evaluated in IoT and IoMT systems.

Continuing with the shift toward ML-driven IDS approaches, early IDS explored the efficacy of algorithms such as Decision Trees and SVMs, both of which achieve promising results when tested on benchmark datasets. For example, a Hybrid Decision Tree tested and evaluated on the NSL-KDD dataset achieves 83% accuracy, outperforming other ML methods such as SVM and KNN [6]. However, this model does not distinguish between attack types such as DoS, MITM, and others, which is desirable for IoMT security; the authors suggest future work addresses this limitation.

Deep learning methods provide a solution to this, a deep reinforcement learning model was developed and tested alongside traditional ML methods to detect malicious network traffic in IoMT networks. Using the NSL-KDD, the best performing model was the Double Deep Q-Network (DDQN) achieving an accuracy of around 80% and F1 score of 0.91[7]. Comparing this the Hybrid Decision Tree, it is clear that this form of deep reinforcement learning excels in attack detection. A noticeable drawback of deep learning here is the time taken to train the model compared to traditional ML as well as the lack of interpretability it provides.

2.3 Explainable Artificial Intelligence (XAI) in Network Security

Explainable Artificial Intelligence (XAI) refers to a set of tools which aim to make the predictions made by a deep learning model understandable to humans, this is in the form of showing the features the model decides played the biggest role in making the final prediction. Obviously, in a healthcare setting, the ability to trust is essential with the IDS system securing an IoMT network because we do not want to risk genuine medical communications being interrupted, which could lead to disruptions to patients treatment. However, we still want to accurately prevent malicious attacks that could cause reduced patient safety and sensitive medical data being compromised. Therefore, integrating XAI into a deep learning IDS allows the healthcare industry to understand and validate the predictions of the model. The two main types of XAI models are model-specific and model-agnostic. Model specific XAI is suited to a specific group of models, meaning they offer a deep insight into the models decision making, however they cannot be applied to a wide variety of model types. Model agnostic techniques are the opposite, they can be applied to many model types with a less in-depth insight. [8]. Some examples of XAI techniques are LIME and SHAP. LIME (Local Interpretable Model-agnostic Explanations) is an algorithm proposed in 2016, it explains the prediction of a classifier (model-agnostic) though approximating it with a simpler and interpretable model [9]. Later, SHAP (Shapley Additive exPlanations) was introduced in 2017 [10], similar to LIME, it is a model-agnostic algorithm that determines how each feature contributes to a models output

3 Project Specification

3.1 Motivation

The use of IoMT systems continues to spread in the field of modern healthcare, with many routine health checks now able to be performed remotely and continuously. Unfortunately, with this increase of its use, there has also been an increase in the amount of cyber-attacks focused on healthcare services [2]. The main goal and motivation of this project is to address the lack of trust that can be placed in Machine Learning based Intrusion Detection Systems, through developing an explainable model whose decisions can be verified and understood.

3.2 Objectives

The primary objectives I will follow this project are outlined as followed:

- Review current literature on AI-based network attack detection within IoT and IoMT networks
- Develop an accurate and interpretable attack detection model and select a suitable XAI technique
- Evaluate the models performance using the CIC2024 IoMT dataset
- Assess the models interpretability and its suitability for cybersecurity specifically to a healthcare context

References

- [1] K. Baseer, K. Sivakumar, D. Veeraiah, G. Chhabra, P. K. Lakineni, M. J. Pasha, R. Gandikota, and G. Harikrishnan, “Healthcare diagnostics with an adaptive deep learning model integrated with the internet of medical things (iomt) for predicting heart disease,” *Biomedical Signal Processing and Control*, vol. 92, p. 105988, 2024.
- [2] S. Alder. (2025) Healthcare data breach statistics. The HIPAA Journal. Accessed: October 27, 2025. [Online]. Available: <https://www.hipaajournal.com/healthcare-data-breach-statistics/>
- [3] S. Dadkhah, E. C. P. Neto, R. Ferreira, R. C. Molokwu, S. Sadeghi, and A. A. Ghorbani, “Ciciomt2024: A benchmark dataset for multi-protocol security assessment in iomt,” *Internet of Things*, vol. 28, p. 101351, 2024.
- [4] Z. Almulla, H. Almajed, and M. Rahman, “A layered security perspective on internet of medical things: Challenges, risks, and technological solutions.” *International Journal of Advanced Computer Science & Applications*, vol. 16, no. 5, 2025.
- [5] K. A. Da Costa, J. P. Papa, C. O. Lisboa, R. Munoz, and V. H. C. de Albuquerque, “Internet of things: A survey on machine learning-based intrusion detection approaches,” *Computer Networks*, vol. 151, pp. 147–157, 2019.
- [6] S. M. Taghavinejad, M. Taghavinejad, L. Shahmiri, M. Zavvar, and M. H. Zavvar, “Intrusion detection in iot-based smart grid using hybrid decision tree,” in *2020 6th International Conference on Web Research (ICWR)*. IEEE, 2020, pp. 152–156.
- [7] M. Lopez-Martin, B. Carro, and A. Sanchez-Esguevillas, “Application of deep reinforcement learning to intrusion detection for supervised problems,” *Expert Systems with Applications*, vol. 141, p. 112963, 2020.
- [8] K. Devireddy, “A comparative study of explainable ai methods: Model-agnostic vs. model-specific approaches,” *arXiv preprint arXiv:2504.04276*, 2025.
- [9] M. T. Ribeiro, S. Singh, and C. Guestrin, “‘‘ why should i trust you?’’ explaining the predictions of any classifier,” in *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*, 2016, pp. 1135–1144.
- [10] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” *Advances in neural information processing systems*, vol. 30, 2017.